A classical H^2 obstruction, read as supply-chain inventory: the inventory gerbe and its band

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Abstract

A supply chain, abstracted to a small category $\mathcal{C}$, carries inventory as local stock-data glued along logistics operations. When charts are compared by *equality* of counts, consistency is a degree-one class $[\theta_R] \in H^1(N(\mathcal{C}); R)$ (companion note). Here we refine the comparison to *isomorphism of stock-states*: lots are fungible, so charts match only up to a relabelling group $K = \mathrm{Aut}(F)$, and the gluing data becomes an $\mathrm{Aut}F$-gerbe over the nerve $N(\mathcal{C})$. The resulting obstruction is *not new mathematics*: it is the classical central-extension lifting class on $N(\mathcal{C})$, read for the first time as an inventory invariant. We make this precise and honest. The obstruction is *staged* after Giraud: a global inventory exists iff *two* obstructions vanish — first the *band* $\beta \in H^1(N(\mathcal{C}); \mathrm{Out}(\mathrm{Aut}F))$, a genuinely nonabelian datum, and then the *gerbe class* $[\mathfrak{g}] \in H^2(N(\mathcal{C}); Z(\mathrm{Aut}F))$ (Corollary 18). The gerbe class vanishes on every 1-dimensional (free / DAG / tree) chain, and is nonzero already on a single reversible loop $B(\mathbb{Z}/2)$ — but only for a *nonabelian* fibre with nontrivial centre — and, for the n-lot ring $\mathrm{Aut}F = D_n$, sharply, exactly when $4 \mid n$, the six-lot ring $D_6 \cong \mathbb{Z}/2 \times S_3$ being nonabelian with centre yet always rectifiable over the trivial-band stratum (Theorems 22, 28; a nontrivial band obstructs independently, Remark 30). It is logically independent of the Zappa–Szép weld $[\omega]$, not because the two live over different coefficients — on $B(\mathbb{Z}/2)$ they can lie in the *same* group and even coincide — but because $[\mathfrak{g}]$ depends on the inventory datum and $[\omega]$ does not (Theorem 24). Finally, both strata are invisible to plain fungibility: $\mathrm{Aut}F = S_n$ is centreless *and* complete for $n \geq 3$, $n \neq 6$, so the two-tier phenomenon is a feature of *gauge/torsor-structured* stock, not of interchangeable commodities (Theorem 27). The result places inventory consistency as a fourth distinct H^2 home on $\mathrm{DCont} \simeq \mathsf{Cat}$, distinguished by its *input datum*.

1 Introduction

Composition is where correctness is won or lost, and inventory is where the price is paid. When two delivery routes reconverge on the same stock-keeping unit, when stock is moved and merged

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and counted again downstream, the question is always whether the local records glue to one consistent global record. When they do not, the failure has an economic name: phantom inventory, a reconciliation that never closes, a write-off. A compositional theory of supply chains earns its keep by telling you *which* such failures can occur and *where* they live.

The categorical model is standard. A supply chain — nodes as objects, admissible logistics operations as morphisms — is a small category $\mathcal{C}$; equivalently, by Ahman and Uustalu [1], a *directed container*. Its nerve $N(\mathcal{C})$ is the simplicial set whose n-simplices are composable strings of operations, and the cohomology of $N(\mathcal{C})$ is the natural home for obstructions to gluing local data along $\mathcal{C}$.

In a companion note [2] we asked when inventory, recorded as counts transported along operations, glues consistently. Writing the inventory as a resource presheaf R on $\mathcal{C}$, the answer was clean: consistency is the vanishing of a *degree-one* class $[\theta_R] \in H^1(N(\mathcal{C}); R)$, the route-reconvergence discrepancy of stock counts, and this class is logically independent of the Zappa–Szép weld obstruction $[\omega] \in H^2(\mathrm{Sk}_{\mathcal{C}}; \mathcal{D})$ that governs whether $\mathcal{C}$ factors as a product of two chains at all [3, 4]. That note flagged one refinement as the natural next question, and the present note answers it.

The gap. Equality of counts is the wrong comparison when lots are *fungible*. Two pallets of the same grade are interchangeable; a local stock-state is a labelled configuration, but the labels carry no information, so two charts should be regarded as matching when they are *isomorphic*, not when they are literally equal. Each inventory fibre F then carries a *relabelling group* $K := \mathrm{Aut}(F)$, and comparison of local charts is canonical only up to K. Gluing up-to-isomorphism is not a gluing-of-sets problem; it is a gluing-of-*stacks* problem — a gerbe over $N(\mathcal{C})$. The companion note anticipated that the obstruction would rise to H^2, but left three questions open. Does the obstruction equal $[\omega]$, which also lives in H^2? What exactly must vanish for a global inventory to exist? For which fibres is the phenomenon even nonvacuous?

What is new, and what is not. We settle all three. The honest framing matters, and we state it at the outset: the obstruction is *classical mathematics*. It is the central-extension lifting class of Schreier and Eilenberg–MacLane, specialised to the nerve $N(\mathcal{C})$ and pulled back along the relabelling transport; for small categories this is the extension theory of Ho [11] and Wells [12], classified cohomologically by Baues–Wirsching [13]. An arXiv search (October 2026) found no prior reading of it as an inventory invariant, so the *application* is new; the *mathematics* is not. We pitch it accordingly — a classical obstruction, newly read as inventory — and the work of the note is to get the reading exactly right.

The results. Fix a fibre F with relabelling group $K = \mathrm{Aut}(F)$, centre $Z := Z(K)$, inner group $\mathrm{Inn}K = K/Z$, and outer group $\mathrm{Out}K$. Two sequences organise everything:

$$1 \longrightarrow Z \longrightarrow K \xrightarrow{\pi} \mathrm{Inn}K \longrightarrow 1 \qquad \text{(central extension of } K\text{)} \qquad (\star)$$

$$1 \to Z \to K \xrightarrow{\mathrm{conj}} \mathrm{Aut}K \xrightarrow{q} \mathrm{Out}K \to 1, \quad \mathrm{Inn}K = \ker q \qquad \text{(inner/outer tower)} \qquad (\star\star)$$

The gluing datum of fungible inventory is an $\mathrm{Aut}F$-gerbe over $N(\mathcal{C})$ whose structure group is K. Classifying it is *staged*, exactly as in Giraud's non-abelian cohomology: the *band* $\beta \in H^1(N(\mathcal{C}); \mathrm{Out}K)$ — the outer part of the relabelling transport — must vanish before the degree-two class is even defined, and then that class $[\mathfrak{g}] \in H^2(N(\mathcal{C}); Z)$ is the lifting obstruction of $(\star)$. Our results are:

Theorem 1 (Rectification $\Leftrightarrow$ vanishing; Theorem 12)**.** *Once the band is trivial, a global inventory — a strict functor $\mathcal{C} \to BK$ realising the up-to-iso gluing — exists if and only if $[\mathfrak{g}] = 0$ in*

$H^2(N(\mathcal{C}); Z)$. *This is the Schreier / Eilenberg–MacLane central-extension lifting criterion, specialised to* $N(\mathcal{C})$.

Corollary 2 (The staged obstruction; Corollary 18)**.** *A global inventory exists if and only if* both *strata vanish:* (i) *the band* $\beta = 0 \in H^1(N(\mathcal{C}); \mathrm{Out}K)$ *and* (ii) *the gerbe class* $[\mathfrak{g}] = 0 \in H^2(N(\mathcal{C}); Z)$*, the latter well-defined only once* (i) *holds.*

Theorem 3 (Degree-genuineness; Theorem 22)**.** *If* $|N(\mathcal{C})|$ *is homotopy equivalent to a* 1-dimensional *complex — in particular if* $\mathcal{C}$ *is the free category on a quiver (a tree-like or DAG chain) — then* $H^{\geq 2}(N(\mathcal{C}); Z) = 0$*, so* $[\mathfrak{g}] = 0$ *for every fibre. There is a minimal one-node witness,* $\mathcal{C} = B(\mathbb{Z}/2)$ *with the fibre a four-lot ring (*$K = \mathrm{Aut}F = D_4$*), for which* $[\mathfrak{g}] \neq 0$*.*

Theorem 4 (Independence from the weld; Theorem 24)**.** $[\mathfrak{g}] \in H^2(N(\mathcal{C}); Z)$ *and the Zappa–Szép weld* $[\omega] \in H^2(\mathrm{Sk}_{\mathcal{C}}; \mathcal{D})$ *are logically independent: over the family of pairs (inventory datum, weld datum) on a fixed* $\mathcal{C}$*, all four vanishing patterns occur. The mechanism is* not *a difference of coefficients — on* $\mathcal{C} = B(\mathbb{Z}/2)$ *the two classes can lie in the same group* $H^2(\mathbb{Z}/2; \mathbb{Z}/2)$ *and even coincide — but a difference of* input data*:* $[\mathfrak{g}]$ *depends on the inventory fibre,* $[\omega]$ *does not.*

Theorem 5 (Nonvacuousness; Theorem 27)**.** $[\mathfrak{g}]$ *can be nonzero only if* K *is* nonabelian with nontrivial centre *(*$\mathrm{Inn}K \neq 1$ *and* $Z \neq 1$*); the minimal such* K *are* D_4 *and* Q_8*. This necessary condition is* not *sufficient, and for the canonical fibre the sharp criterion is arithmetic: for* n *lots in a ring (*$K = \mathrm{Aut}F = D_n$*) the gerbe is essentially nonzero iff the central extension* ($\star$) *is non-split iff* $4 \mid n$ *— so the six-lot ring* $D_6 \cong \mathbb{Z}/2 \times S_3$ *is nonabelian with nontrivial centre yet always rectifiable over the trivial-band stratum (Theorem 28; the banded case is a genuine exception, Remark 30). For* F *a bare* n*-element set of indistinguishable lots,* $\mathrm{Aut}F = S_n$ *is abelian or centreless for every* n *and complete (so* $\mathrm{Out} = 1$*) for* $n \geq 3$*,* $n \neq 6$*: fungible stock carries* neither *obstruction. The two-tier phenomenon is a feature of gauge/torsor-structured stock, not of interchangeable commodities.*

The main results are visible by page two. The picture is a *two-tier* obstruction with a *staged* degree-two tier:

tier	charts compared by	obstruction	coefficients, degree
1	equality of counts	$[\theta_R]$	R (resource group), H^1
2a (band)	iso of states, outer part	β	$\mathrm{Out}(\mathrm{Aut}F)$ (nonabelian), H^1
2b (gerbe)	iso of states, central part	$[\mathfrak{g}]$	$Z(\mathrm{Aut}F)$ (fibre centre), H^2
(precond.)	does $\mathcal{C}$ factor at all	$[\omega]$	$\mathcal{D}$ (weld vertex groups), H^2

The method. One classical fact, specialised to $N(\mathcal{C})$, does the work. Comparing inventory charts across an operation transports the relabelling group K by an automorphism, functorial only up to inner automorphism; its outer reduction is the band β (Lemma 16), and $\beta = 0$ iff the transport is inner-valued up to relabelling (Theorem 17). When $\beta = 0$ the transport descends to a functor $\rho : \mathcal{C} \to B(\mathrm{Inn}K)$, and the gerbe is its *gerbe of lifts* through ($\star$): choosing set-section lifts $T_f = s(\rho_f)$ exhibits a Z-valued 2-cocycle γ whose class is $[\mathfrak{g}]$ (§3). Rectification then reads as trivialising γ by a central 1-cochain — an elementary coboundary computation (Theorem 12). The remaining theorems are a simplicial computation on the nerve (Theorem 22), a comparison with explicit witnesses (Theorem 24), and the classical automorphism arithmetic of S_n (Theorem 27).

Consequences and placement. Three points up front. (i) The refinement is *exactly* one categorical level: equality-gluing is H^1, iso-gluing is H^2, with a nonabelian H^1 band in between that must vanish first. (ii) Because $[\mathfrak{g}]$ and $[\omega]$ can share a group, the temptation to identify "inventory is inconsistent up to relabelling" with "the chain fails to weld" is a genuine trap — they can even be equal — and only the *input datum* separates them; Theorem 24 makes this precise. (iii) The nonvacuousness criterion sharpens the applications story: the economically relevant tier-two case is not plain fungibility of many interchangeable units (centreless, complete S_n) but *gauge-valued* stock — allocation defined up to a group action — whose automorphism group is nonabelian with a centre. Within the Kodamai program, $[\mathfrak{g}]$ is the *fourth* distinct H^2 home on $\mathrm{DCont} \simeq \mathsf{Cat}$, after the Zappa–Szép holonomy, Ferri prunability, and the Baues–Wirsching class H^2_{Cat}; it is distinguished from them by its *input datum* (the inventory fibre), not by a coefficient-system discriminator.

Outline. Section 2 recalls supply chains as directed containers, the two sequences ($\star$)–($\star\star$), and the Giraud strata we use. Section 3 defines the object as the gerbe of lifts and exhibits its cocycle. Section 4 proves the rectification criterion (Theorem A) and identifies it as classical. Section 5 proves the band criterion and the staged corollary (the band boundary). Section 6 proves degree-genuineness (Theorem B). Section 7 proves independence from the weld (Theorem C). Section 8 proves the nonvacuousness criterion and the fungibility dichotomy (Theorem D). Section 9 states the scope, the honest open problem (coupled strata), and the two-tier bottom line.

Note on this revision. This note corrects the version of 9 October 2026 (commit `9eb0941`) following Rick's referee report of 10 October. The corrections are substantive and we state them openly rather than paper over them: the earlier degree-genuineness hypothesis ("homotopy 1-truncated") was false and is replaced by "homotopy equivalent to a 1-complex" (§6); the earlier minimal witness ($K = \mathbb{Z}/2$) was wrong — a bare set carries no tier-two class — and is replaced by the four-lot ring $K = D_4$ (§6, 8); the earlier "different coefficients" explanation of independence was false and is replaced by the input-data mechanism (§7); and the band, previously hypothesised away, is now settled (§5). The gerbe cocycle is renamed γ to free the symbol ω for the weld.

A second round of corrections (11 October 2026) follows Rick's round-2 report. **N1:** the proof of Corollary 18 had silently used a definition of global inventory (a relabelling μ trivialising α outright) different from the one in Definition 6/Theorem 12, and was *false* under it ($\mathcal{C} = B(\mathbb{Z}/2)$, $K = D_4$, $\alpha(g) = \mathrm{conj}(s)$: $\beta = 0$ and $[\mathfrak{g}] = 0$, yet no μ trivialises α). We now fix a single definition (D2), make explicit the standing strictness / 2-cell hypothesis that the inner reduction $\alpha = \mathrm{conj} \circ \rho$ requires (§2.3), and reframe the corollary as the classical two-stage crossed-module obstruction (Remark 19). **N2:** the biconditional "$[\mathfrak{g}] \neq 0 \iff K$ nonabelian with nontrivial centre" — which appeared in the registry and cover email, though not in the printed Theorem 27 (whose "only if" is correct) — is false; the honest witness is $D_6 \cong \mathbb{Z}/2 \times S_3$. We retract it and add the sharp criterion (Theorem 28, §8.1): for the n-lot ring, $[\mathfrak{g}]$ is essentially nonzero iff $4 \mid n$.

2 Preliminaries

The section records only what the proofs use. A reader fluent in directed containers and non-abelian cohomology can skim to Section 3.

2.1 Supply chains as directed containers and their nerve

A supply chain is modelled as a small category $\mathcal{C}$: objects are nodes (warehouses, plants, stock locations), morphisms are admissible logistics operations, composition is sequencing. By Ahman–

Uustalu [1] a small category is the same thing as a directed container, so the model lives natively in the container program.

The *nerve* $N(\mathcal{C})$ is the simplicial set with $N(\mathcal{C})_0 = \mathrm{Ob}\mathcal{C}$, $N(\mathcal{C})_1 = \mathrm{Hom}\mathcal{C}$, and $N(\mathcal{C})_n$ the set of composable strings $c_0 \xrightarrow{f_1} c_1 \to \cdots \xrightarrow{f_n} c_n$. For an abelian coefficient group A with (here) trivial $\mathcal{C}$-action, $H^\bullet(N(\mathcal{C}); A)$ is the cohomology of the normalised cochain complex with the usual alternating-face differential. Two facts are used below:

- For a one-object category $\mathcal{C} = BG$ (a group G viewed as a category), $N(BG)$ is the standard classifying space and $H^\bullet(N(BG); A) = H^\bullet(G; A)$ is group cohomology.
- If $|N(\mathcal{C})|$ is homotopy equivalent to a 1-dimensional CW complex then $H^{\geq 2}(N(\mathcal{C}); A) = 0$. This holds for the free category $F(Q)$ on a quiver Q, whose classifying space is homotopy equivalent to the graph $|Q|$.

We emphasise the second fact in its correct form: it is *homotopy* 1*-dimensionality*, not homotopy 1-truncatedness, that forces $H^{\geq 2} = 0$. A $K(\pi, 1)$ such as $B(\mathbb{Z}/2)$ is 1-truncated yet has $H^{\geq 2} \neq 0$; see Remark 23.

2.2 Inventory fibres and the two sequences

An *inventory fibre* F is the local type of a stock-state at a node: a configuration of lots. Fungibility means the labels carry no information, so the relevant symmetry is the *relabelling group*

$$K := \mathrm{Aut}(F), \qquad Z := Z(K) \text{ (centre)}, \qquad \mathrm{Inn}K := K/Z \text{ (inner group)},$$

fitting into the central extension ($\star$). We write $\pi : K \to \mathrm{Inn}K$ for the quotient and reserve $s : \mathrm{Inn}K \to K$ for a chosen *set-section* with $s(e) = e$ (not a homomorphism in general). The band lives in the outer group $\mathrm{Out}K$ of the inner/outer tower ($\star\star$); $\mathrm{Inn}K = \ker q \cong K/Z$ is the image of conjugation.

2.3 The gerbe, its band, and the Giraud strata

Comparing two inventory trivialisations of the same fibre across an operation f identifies them only up to the relabelling group K, i.e. transports K by an *automorphism* $\alpha_f \in \mathrm{Aut}K$. Re-trivialising a chart conjugates the transport, so α is a priori functorial only up to inner automorphisms, $\alpha_{gf} = \mathrm{conj}(k_{g,f}) \cdot \alpha_g \alpha_f$ with witnessing *comparison data* $k_{g,f} \in K$. **Throughout we take trivial comparison data,** $k_{g,f} \equiv e$: the transport is then a *strict* functor $\alpha : \mathcal{C} \to B(\mathrm{Aut}K)$ into the 1-category $B(\mathrm{Aut}K)$. (In general, re-trivialising charts makes α functorial only up to the inner slack recorded by the $k_{g,f}$, i.e. a *pseudofunctor* $\alpha : \mathcal{C} \to B(\mathrm{Aut}K)$ valued in the automorphism 2-group $\mathrm{AUT}(F)$; Remark 11.) Under this standing hypothesis a strict inner-valued α is itself a genuine functor to $B(\mathrm{Inn}K)$, so the degree-two class is well defined. *Scope of the sharp results.* The sharp n-lot criterion of §8.1 is stated for *trivial* comparison data, $k_{g,f} \equiv e$, so that $[\mathfrak{g}] = \rho^*[\star]$. A nontrivial *central* comparison datum $k_{g,f} \in Z$ is itself a 2-cocycle and contributes a class $[k] \in H^2(N(\mathcal{C}); Z)$; the total class becomes $[\mathfrak{g}] = \rho^*[\star] + [k]$, and the sharp divisibility line then reads $2 \mid n$ rather than $4 \mid n$ (Remark 11, §8.1). All results below are stated under $k_{g,f} \equiv e$. Collecting the local trivialisations into a stack makes the gluing datum an $\mathrm{Aut}F$-gerbe over $N(\mathcal{C})$: a stack of groupoids locally equivalent to $B(\mathrm{Aut}F)$ (Giraud [9], III; Breen [10] for the bitorsor/cocycle presentation). A *global inventory* realises the up-to-iso gluing by a *strict* relabelling of charts; we make this precise, in gauge-invariant form, in Definition 6. (Following Giraud, the band is *part of the data* of the

gerbe, not an obstruction to its neutrality; a global inventory is the vanishing of the *staged lifting obstruction* of Corollary 18, not a "section realised on the nose".)

Giraud's classification of a G-gerbe over a base X (for us $G = \mathrm{Aut}F = K$, $X = N(\mathcal{C})$) factors into *three strata*:

stratum	datum	invariant	home
1 (unbanded)	structure 2-group cocycle	$\pi_0 \cong H^1(X; \mathrm{AUT}\,K)$	nonabelian H^1
2 (band)	outer transport $\rho := q \circ \alpha$	$\beta := [\rho] \in H^1(X; \mathrm{Out}K)$	nonabelian H^1
3 (fixed band)	centre cocycle	$[\mathfrak{g}] \in H^2(X; Z)_\beta$	abelian H^2

We work the classification at strata 2 and 3, which is where a *strict global section* lives: stratum 1 concerns which 2-group presents the gerbe, which for a fixed fibre F and constant relabelling group is already pinned. The **band** β must vanish before the degree-two coefficients Z are untwisted and $[\mathfrak{g}]$ is defined; this staging is the content of Section 5. For stratum 3 we need only one classical fact, imported rather than re-derived: an abelian-band gerbe over $N(\mathcal{C})$ with band A (trivial $\mathcal{C}$-action) is classified by $H^2(N(\mathcal{C}); A)$, the trivial class being the one admitting a global section (Giraud [9], III). Everything else — the cocycle, its class, the rectification algebra, the band criterion — we compute in place.

Conventions for nonabelian H^1. For a group Q (constant coefficients) a *cocycle* is a functor $\mathcal{C} \to BQ$, i.e. a map $\gamma : \mathrm{Mor}\mathcal{C} \to Q$ with $\gamma(gf) = \gamma(g)\gamma(f)$, $\gamma(\mathrm{id}) = e$; two cocycles are *cohomologous* iff related by an object-relabelling $\lambda : \mathrm{Ob}\mathcal{C} \to Q$, $\gamma'(f : x \to y) = \lambda(y)\gamma(f)\lambda(x)^{-1}$. Then $H^1(N(\mathcal{C}); Q) := Z^1/\!\sim$, pointed by $\gamma \equiv e$; $[\gamma] = 0$ means $\gamma \sim e$. For Q abelian this is ordinary simplicial H^1; for Q nonabelian it is a pointed set.

3 The inventory gerbe as the gerbe of lifts

We define the object directly as the *gerbe of lifts* of the inner transport through the central extension ($\star$). This is the honest construction: it makes transparent that the degree-two class is extra lifting data pulled back along the transport, not a number computed from the fibre F in isolation.

Definition 6 (The inventory gerbe)**.** Fix a fibre F with $K = \mathrm{Aut}F$, let $\alpha : \mathcal{C} \to B(\mathrm{Aut}K)$ be the (strict; §2.3) relabelling transport, and suppose the band is trivial ($\beta = 0$; Section 5). Then a gauge $\mu : \mathrm{Ob}\mathcal{C} \to \mathrm{Aut}K$ reduces α to an inner-valued transport; because α is strict, this reduced transport is itself a *genuine functor*

$$\rho : \mathcal{C} \longrightarrow B(\mathrm{Inn}K), \qquad \mathrm{Inn}K = K/Z,$$

with $\mathrm{conj} \circ \rho = \mu \cdot \alpha$. (It is here that strictness is used: a strict inner-valued α is itself a functor to $B(\mathrm{Inn}K)$, so ρ — hence $[\mathfrak{g}]$ — is well defined.) The *inventory gerbe* is the Z-gerbe of lifts of ρ through π in ($\star$): its local objects are lifts of ρ_f to K, and its band is the centre Z with trivial $\mathcal{C}$-action. A *global inventory* is a pair (μ, T) of a gauge $\mu : \mathrm{Ob}\mathcal{C} \to \mathrm{Aut}K$ and a strict functor $T : \mathcal{C} \to BK$ with

$$\mu(y)\,\alpha(f)\,\mu(x)^{-1} = \mathrm{conj}(Tf) \qquad \text{for every } f : x \to y \tag{D2}$$

— equivalently, a strict lift $T : \mathcal{C} \to BK$ of the reduced transport ρ. This is the single notion used throughout; it refines the informal "strict functor $\mathcal{C} \to BK$" by making the gauge μ (hence the reduction $\beta = 0$) explicit.

Remark 7 (Why the gerbe of lifts, and not a bare K-gerbe)**.** It is tempting to call the object a "K-gerbe with trivial band" and invoke the band formalism to reduce its coefficients to Z. That is imprecise: a K-gerbe with trivial band and the Z-gerbe of lifts of ρ are *different* objects — for $\rho = \mathrm{id}$ the inner form $\mathbf{Tors}(\mathrm{Inn}K)$ has a global section while the Z-gerbe of lifts does not — so "global section $\Leftrightarrow [\mathfrak{g}] = 0$" depends on which one is meant. We therefore *define* the object as the gerbe of lifts (equivalently, as the $\mathrm{AUT}(F)$-pseudofunctor of Remark 11), and no separate band-is-the-centre lemma is needed: the coefficients are Z by construction.

3.1 The cocycle

Definition 8 (Section cocycle)**.** Choose set-section lifts $T_f := s(\rho_f) \in K$ for each operation f, with $T_{\mathrm{id}} = e$. Because both $T_g T_f$ and T_{gf} map to $\rho_g \rho_f = \rho_{gf}$ under π, their ratio lands in $\ker \pi = Z$:

$$T_g T_f = \gamma(g, f) \cdot T_{gf}, \qquad \gamma(g, f) \in Z. \tag{$\dagger$}$$

Thus $\gamma = \rho^*(\text{extension 2-cocycle of } (\star))$ is a normalised 2-cochain on $N(\mathcal{C})$ valued in Z.

Lemma 9 (γ is a cocycle and $[\mathfrak{g}]$ is well-defined)**.** *Associativity of multiplication in K, together with the centrality of Z (so conjugation acts trivially on the values of γ), gives the normalised 2-cocycle identity*

$$\gamma(h, g)\, \gamma(hg, f) = \gamma(g, f)\, \gamma(h, gf), \qquad \textit{i.e. } \delta\gamma = 0,$$

and changing the set-section s replaces γ by a cohomologous cochain. Hence $[\mathfrak{g}] := \rho^*[\star] = [\gamma] \in H^2(N(\mathcal{C}); Z)$ is well-defined.

Remark 10 (Gauge-dependence of $[\mathfrak{g}]$)**.** The reducing gauge μ of Definition 6 is pinned only up to an outer twist ($\lambda := q \circ \mu$ may be any $\mathrm{Out}K$-valued object-relabelling trivialising the band), so ρ, and with it γ, are determined only up to the induced $\mathrm{Aut}(Z)$-action on the coefficients. The *vanishing* $[\mathfrak{g}] = 0$ is nevertheless gauge-invariant — it is exactly the existence of a strict lift in (D2) — and that is all Theorem 12 and Corollary 18 use.

Proof. Expand $T_h(T_g T_f) = (T_h T_g) T_f$ using ($\dagger$) three times and cancel the common T_{hgf}; centrality lets the Z-values commute past the T's. Replacing s by s' sets $T'_f = c_f T_f$ with $c_f \in Z$, and ($\dagger$) gives $\gamma'(g, f) = \gamma(g, f)\, (\delta c)(g, f)$. □

Remark 11 (Face 1 is definitional: $[\mathfrak{g}]$ is a k-invariant, not a number from F)**.** When K is abelian, $\mathrm{Inn}K = 1$, ρ is trivial, and the gerbe of lifts of Definition 6 is *trivial*: $[\mathfrak{g}] = 0$. So an abelian fibre carries no inventory class by this construction. The abelian K-gerbes presented by a free 2-cocycle $\gamma \in Z^2(N(\mathcal{C}); K)$ — the "abelian face" of the earlier version — are therefore *not* determined by F: a pseudofunctor into the 1-category BK is just a functor and carries no 2-cell data, so a nonzero class is extra input. Precisely, the datum is a pseudofunctor into the automorphism 2-group $\mathrm{AUT}(F)$ (with $\pi_1 = \mathrm{Out}K$, $\pi_2 = Z$), and $[\mathfrak{g}]$ is its *k-invariant* [15, 16]. We record this honestly: Face 1 is the *definition* of a classification datum, not a computation from the fibre. The constructive, inventory-driven class is the gerbe of lifts of Definition 6, which requires a *nonabelian* K (so that ρ can be nontrivial) — as Theorem 27 makes precise.

The sharp arithmetic of §8.1 assumes the comparison data is trivial, $k_{g,f} \equiv e$, so that $[\mathfrak{g}] = \rho^*[\star]$. If instead $k_{g,f}$ is central and nontrivial ($k_{g,f} \in Z$), it is a 2-cocycle in its own right and contributes a class $[k] \in H^2(N(\mathcal{C}); Z)$; the pseudofunctor's class is then $[\mathfrak{g}] = \rho^*[\star] + [k]$. A central k can supply a nonzero class whenever $Z \neq 1$ — for the n-lot ring, whenever n is even — so with central comparison data allowed the sharp line below reads $2 \mid n$ rather than $4 \mid n$. The results of this paper are stated under $k_{g,f} \equiv e$.

4 Rectification is the vanishing of $[\mathfrak{g}]$

Theorem 12 (Theorem A)**.** *With the band trivial, a global inventory — a strict functor $\mathcal{C} \to BK$ lifting ρ — exists if and only if $[\mathfrak{g}] = 0$ in $H^2(N(\mathcal{C}); Z)$, where $Z = Z(\mathrm{Aut}F)$.*

Proof. Re-choosing the lift at each operation f multiplies T_f by an element $c_f \in Z$ (it moves T_f within its Z-torsor of lifts of ρ_f). A candidate global inventory is a 1-cochain $c : \mathrm{Mor}\mathcal{C} \to Z$ producing new transitions $T'_f = c_f T_f$, and strictness is $T'_{gf} = T'_g T'_f$. Using (†) and centrality,

$$c_{gf}\, T_{gf} = T'_{gf} \overset{!}{=} T'_g T'_f = c_g c_f\, T_g T_f = c_g c_f\, \gamma(g,f)\, T_{gf},$$

which holds iff $\gamma(g,f) = c_{gf}\, c_g^{-1} c_f^{-1} = (\delta c^{-1})(g,f)$. Hence a global inventory exists iff γ is a Z-coboundary, i.e. iff $[\mathfrak{g}] = [\gamma] = 0$. □

Remark 13 (This is the classical lifting criterion)**.** Theorem 12 is *not* new. It is the Schreier / Eilenberg–MacLane criterion for lifting a map into a quotient $\mathrm{Inn}K$ through a central extension (⋆), here pulled back along ρ and specialised to the simplicial site $N(\mathcal{C})$: for small categories this is the extension theory of Ho [11] and Wells [12], with the cohomological classification of Baues–Wirsching [13]. We present it as a specialisation so read, not as a theorem of this note; the contribution is the *reading* of ρ, Z, and $[\mathfrak{g}]$ as inventory transport, fibre centre, and rectification obstruction.

Remark 14 (Concrete check: D_4 and Q_8)**.** Take $\mathcal{C} = B((\mathbb{Z}/2)^2)$ and $K \in \{D_4, Q_8\}$, each with $Z = \mathbb{Z}/2$, $\mathrm{Inn}K = (\mathbb{Z}/2)^2$, and $\rho = \mathrm{id}$ (the inner transport is the full inner group). A strict lift would be a splitting homomorphism $(\mathbb{Z}/2)^2 \to K$ of (⋆); none exists, since K is nonabelian while $(\mathbb{Z}/2)^2$ is abelian. Correspondingly the section-cocycle γ of (†) is not a coboundary, so $[\mathfrak{g}] \neq 0$ and rectification fails. The non-splitting and $[\mathfrak{g}] \neq 0$ are confirmed by brute-force enumeration over $\mathbb{F}_2$ (`inventory_gerbe_h2.py`; the reduced-nerve H^2 engine of the Lean witness file [6]).

5 The band, and the staged obstruction

The earlier version hypothesised the band away. We now settle it, following Giraud's three-stage factorisation over the simplicial site $N(\mathcal{C})$. This is the program's first genuinely *nonabelian* content: the band lives in H^1 with the nonabelian coefficient *group* $\mathrm{Out}K$.

Definition 15 (Band)**.** The *band* (lien) of the inventory gerbe is the outer reduction of the relabelling transport,

$$\rho_{\mathrm{out}} := q \circ \alpha : \mathrm{Mor}\mathcal{C} \longrightarrow \mathrm{Out}K,$$

where $\alpha : \mathrm{Mor}\mathcal{C} \to \mathrm{Aut}K$ is the transport (functorial up to inner) and q is as in (⋆⋆). Its class is $\beta := [\rho_{\mathrm{out}}] \in H^1(N(\mathcal{C}); \mathrm{Out}K)$.

Lemma 16 (The band is always a strict cocycle)**.** *ρ_{out} is a genuine functor $\mathcal{C} \to B(\mathrm{Out}K)$, so β is well-defined regardless of the up-to-inner slack in α.*

Proof. Write $\alpha_{gf} = \iota \cdot \alpha_g \alpha_f$ with $\iota \in \mathrm{Inn}K = \ker q$. Apply q: $\rho_{\mathrm{out}}(gf) = q(\iota)\, q(\alpha_g) q(\alpha_f) = \rho_{\mathrm{out}}(g)\rho_{\mathrm{out}}(f)$, since $q(\iota) = e$. Normalisation is immediate; the inner slack dies in $\mathrm{Out}K$. □

Theorem 17 (Band-triviality criterion; Theorem (a))**.** *$\beta = 0 \in H^1(N(\mathcal{C}); \mathrm{Out}K)$ if and only if the (strict; §2.3) transport α is, up to an $\mathrm{Out}K$-relabelling of objects, inner-valued: iff there is a cocycle $\tilde{\alpha} : \mathcal{C} \to B(\mathrm{Aut}K)$, cohomologous to α via an $\mathrm{Aut}K$-valued object relabelling, with $\tilde{\alpha}(f) \in \mathrm{Inn}K$ for*

every f. (Strictness makes α a genuine cocycle, so "cohomologous" is literal; absent it, read the criterion as "α can be gauged to be inner-valued".) In particular, if α already lands in $\mathrm{Inn}K$ *then $\beta = 0$; this is the literal "trivial band", and the theorem says $\beta = 0$ is its gauge-invariant closure.*

Proof. ($\Leftarrow$) Let $\tilde{\alpha} = \mu \cdot \alpha$ with $\mu : \mathrm{Ob}\mathcal{C} \to \mathrm{Aut}K$ and $\tilde{\alpha}(f) \in \mathrm{Inn}K = \ker q$ for all f. Set $\lambda := q \circ \mu : \mathrm{Ob}\mathcal{C} \to \mathrm{Out}K$. For $f : x \to y$,

$$\rho_{\mathrm{out}}(f) = q(\alpha(f)) = q\big(\mu(y)^{-1}\tilde{\alpha}(f)\mu(x)\big) = \lambda(y)^{-1}\, q(\tilde{\alpha}(f))\, \lambda(x) = \lambda(y)^{-1}\lambda(x),$$

so λ trivialises ρ_{out} and $\beta = 0$.

($\Rightarrow$) If $\beta = 0$ there is $\lambda : \mathrm{Ob}\mathcal{C} \to \mathrm{Out}K$ with $\rho_{\mathrm{out}}(f) = \lambda(y)^{-1}\lambda(x)$. As q is surjective, choose lifts $\mu(x) \in \mathrm{Aut}K$ with $q(\mu(x)) = \lambda(x)$, and set $\tilde{\alpha}(f) := \mu(y)\alpha(f)\mu(x)^{-1}$. Then $q(\tilde{\alpha}(f)) = \lambda(y)\rho_{\mathrm{out}}(f)\lambda(x)^{-1} = e$, so $\tilde{\alpha}(f) \in \mathrm{Inn}K$; and $\tilde{\alpha}$ is a cocycle cohomologous to α via μ. □

This is Giraud stratum 2 specialised to $N(\mathcal{C})$: the band vanishes exactly when the transport is inner-valued up to coboundary. The proof is elementary and Schreier-style, using only the tower ($\star\star$) and the nerve's H^1. For $K = D_4$ (Aut $= D_4$ of order 8, Inn $= (\mathbb{Z}/2)^2$, Out $= \mathbb{Z}/2$ — both parts non-trivial) over $\mathcal{C} = B(\mathbb{Z})$ the criterion holds for all 8 automorphisms (`scratch/clause_a_D4_check.py`).

Corollary 18 (The full obstruction is staged; Corollary (a′))**.** *A global inventory exists if and only if* both (i) $\beta = 0$ *(band trivial, Theorem 17) and* (ii) $[\mathfrak{g}] = 0 \in H^2(N(\mathcal{C}); Z)$ *(the gerbe class, defined once (i) holds). The obstruction is genuinely staged:* $[\mathfrak{g}]$ *is defined only after $\beta = 0$, otherwise the stratum-3 coefficients are β-twisted. Thus Theorem 12 is exactly the band-trivial stratum, and Theorem 17 supplies the prior stratum.*

Proof. Unwind Definition 6: a global inventory is a pair (μ, T) satisfying (D2), with $\mu : \mathrm{Ob}\mathcal{C} \to \mathrm{Aut}K$ and $T : \mathcal{C} \to BK$ strict. *Necessity of (i):* applying q to (D2), the right-hand side $\mathrm{conj}(Tf)$ is inner, so $q(\mu(y)\,\alpha(f)\,\mu(x)^{-1}) = e$, i.e. the gauge μ trivialises the outer transport ρ_{out}; hence $\beta = 0$. *Sufficiency given (i):* Theorem 17 supplies a gauge μ for which $\tilde{\alpha} := \mu \cdot \alpha$ is inner-valued. Because α is a *strict* functor carrying its comparison data $k_{g,f}$ (§2.3), $\tilde{\alpha}$ is a strict functor landing in $\mathrm{Inn}K$, hence $\tilde{\alpha} = \mathrm{conj} \circ \rho$ for a *genuine* functor $\rho : \mathcal{C} \to B(\mathrm{Inn}K)$ — this is the step that needs strictness: a strict inner-valued $\tilde{\alpha}$ is itself a functor to $B(\mathrm{Inn}K)$. The gerbe is then the gerbe of lifts of ρ through ($\star$), and by Theorem 12 a strict T with (D2) exists iff $[\mathfrak{g}] = 0$. So (i)$\wedge$(ii) is necessary and sufficient. □

Remark 19 (Corollary 18 is the classical two-stage obstruction)**.** Corollary 18 is, stripped of the inventory reading, the obstruction to lifting the transport α through the crossed module $K \xrightarrow{\mathrm{conj}} \mathrm{Aut}K$ — equivalently, through the Postnikov tower of the automorphism 2-group $\mathrm{AUT}(F)$, whose homotopy groups are $\pi_1 = \mathrm{Out}K$ and $\pi_2 = Z$. The *primary* obstruction is the band $\beta \in H^1(N(\mathcal{C}); \mathrm{Out}K)$; once it vanishes the *secondary* obstruction is the gerbe class $[\mathfrak{g}] \in H^2(N(\mathcal{C}); Z)$. This staging is classical non-abelian obstruction theory — Eilenberg–MacLane's extensions with non-abelian kernel via a map $\psi : Q \to \mathrm{Out}K$, and Giraud's and Breen's gerbe-theoretic form ([9], IV; [10]). Our contribution is the inventory reading of α, β and $[\mathfrak{g}]$, not the two-stage structure itself.

Remark 20 (The band is where the nonabelian content lives)**.** The band is the first obstruction in the program with a *nonabelian coefficient group* $\mathrm{Out}(\mathrm{Aut}F)$. This is an honest structural statement — genuinely different in kind from the abelian H^2 classes — and it is what the earlier "distinguished by its coefficient system" slogan should have said (that slogan was false for $[\mathfrak{g}]$ versus $[\omega]$; Theorem 24). A nontrivial-band witness, with a clean separation of the two strata:

Proposition 21 (A nonvacuous band; Proposition (c))**.** *Let $K = (\mathbb{Z}/p)^2$ (abelian gauge group of stock) and $\mathcal{C} = B(\mathbb{Z})$ (one reversible/cyclic operation). For abelian K,* $\mathrm{Inn}K = 1$ *and* $\mathrm{Out}K = \mathrm{Aut}K = \mathrm{GL}_2(\mathbb{F}_p)$*, so the band sees the entire transport: $\rho_{\mathrm{out}} = \alpha$. Set $\alpha(t) := A$ for some $A \neq I$. Then*

$$H^1(N(B(\mathbb{Z})); \mathrm{Out}K) = \{\textit{conjugacy classes of } \mathrm{GL}_2(\mathbb{F}_p)\}, \qquad \beta = [A] \neq 0.$$

Moreover the strata separate cleanly: since $N(B(\mathbb{Z})) \simeq S^1$ is 1-dimensional, $H^2(N(\mathcal{C}); Z) = 0$, so the gerbe class is forced *to zero while the band is the sole, nonzero obstruction — living in nonabelian H^1 with coefficients $\mathrm{GL}_2(\mathbb{F}_p)$, prior to and independent of any H^2 story.*

Proof. $N(B(\mathbb{Z})) \simeq S^1$; a cocycle is determined by $\rho_{\mathrm{out}}(t) = A$ and object-relabelling by $g \in \mathrm{Out}K$ sends $A \mapsto gAg^{-1}$, so $H^1 = \{\text{conjugacy classes}\}$ and $\beta = 0$ iff $A = I$. For $p = 2$, $\mathrm{GL}_2(\mathbb{F}_2) \cong S_3$; the transvection $A = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right) \neq I$ gives $\beta \neq 0$. Clean separation: $H^2(S^1; -) = 0$ for any coefficients (`scratch/band_boundary_verify.py`). □

Interpretation. Stock is defined up to a $(\mathbb{Z}/p)^2$-gauge (say two independent mod-p ledger offsets); a cyclic refurbishment process t relabels the gauge by a fixed $\mathrm{GL}_2(\mathbb{F}_p)$ element A. If $A \neq I$, no global choice of gauge is consistent around the loop — a *band* obstruction, an outer twist of the structure group, strictly prior to the degree-two gerbe class. Fungibility can never produce it (Theorem 27); gauge-structured stock does, minimally. Activating both strata at once needs a base with $H^2 \neq 0$ and a nontrivial outer transport (e.g. $\mathcal{C} = B(\mathbb{Z}^2)$, $N \simeq T^2$); the twisted coupling of band to centre on such a base is the natural follow-up and is *not* treated here (Section 9).

6 $[\mathfrak{g}]$ is genuinely a degree-two datum

Theorem 22 (Theorem B)**.** (i) *If $|N(\mathcal{C})|$ is homotopy equivalent to a 1-dimensional complex — in particular if $\mathcal{C}$ is the free category on a quiver (a tree-like or DAG supply chain) — then $H^{\geq 2}(N(\mathcal{C}); Z) = 0$, so $[\mathfrak{g}] = 0$ for every fibre F.* (ii) *There is a minimal one-node witness, $\mathcal{C} = B(\mathbb{Z}/2)$ with the fibre a four-lot ring ($K = \mathrm{Aut}F = D_4$), for which $[\mathfrak{g}] \neq 0$.*

Proof of (i). If $|N(\mathcal{C})| \simeq X$ with X a 1-dimensional CW complex, then $H^{\geq 2}(N(\mathcal{C}); Z) = H^{\geq 2}(X; Z) = 0$. For the free category $F(Q)$ on a quiver Q, $|N(F(Q))| \simeq |Q|$, a graph; hence $H^{\geq 2} = 0$. It is worth seeing this directly on the reconvergent *diamond* $a \rightrightarrows \{b, c\} \rightrightarrows d$ — the paradigmatic two-route chain — whose nerve is *not* literally 1-dimensional: it carries two nondegenerate 2-simplices, the composites $p; r$ and $q; s$ of the two routes, yet $|N|$ is still homotopy equivalent to a graph. A direct simplicial computation over $\mathbb{F}_2$ (`diamond_nerve_H2.py`) gives chain dimensions $(C^0, C^1, C^2) = (4, 6, 2)$, boundary ranks $\mathrm{rk}\, d^0 = 3$, $\mathrm{rk}\, d^1 = 2$, and hence

$$H^0 = 1, \qquad H^1 = 1, \qquad H^2 = 0, \qquad \chi = 4 - 6 + 2 = 0.$$

The lone H^1 is the tier-one reconvergence loop $[\theta_R]$; $H^2 = 0$ says there is no gerbe on a DAG, even one with genuine 2-simplices. □

Proof of (ii). The fibre is four lots arranged in a ring (the 4-cycle); its relabelling group is the dihedral group $K = \mathrm{Aut}F = D_4$ of order 8, with centre $Z = \{1, r^2\} \cong \mathbb{Z}/2$ (the half-turn) and inner group $\mathrm{Inn}K = (\mathbb{Z}/2)^2$. Let $\mathcal{C} = B(\mathbb{Z}/2)$, with reversible operation g ($g^2 = \mathrm{id}$). Choose the inner transport $\rho : B(\mathbb{Z}/2) \to B(\mathrm{Inn}K)$ sending g to $[r] \in \mathrm{Inn}K$ (the class of the rotation; $[r]$ has order 2 since $r^2 \in Z$). Take the set-section lift $T_g := r$ and $T_{\mathrm{id}} := e$. Then (†) gives

$$\gamma(g, g) = T_g T_g T_{g^2}^{-1} = r^2 T_{\mathrm{id}}^{-1} = r^2 \neq 1 \in Z.$$

Every normalised 1-cochain $c : \mathrm{Mor}\mathcal{C} \to Z$ has $(\delta c)(g,g) = c_{g^2} c_g^{-1} c_g^{-1} = c_g^{-2} = 1$ (as $Z = \mathbb{Z}/2$ and $g^2 = \mathrm{id}$), so γ is *not* a coboundary and $[\mathfrak{g}]$ is the generator of $H^2(\mathbb{Z}/2;\mathbb{Z}/2) = \mathbb{Z}/2$. The band is trivial here (ρ is inner-valued, so $\beta = 0$), and `inventory_gerbe_h2.py` / the Lean witness [6] confirm $[\mathfrak{g}] \neq 0$. □

Remark 23 (Why not $K = \mathbb{Z}/2$ — homotopy 1-truncatedness is not enough)**.** The earlier version used the *abelian* fibre $K = \mathbb{Z}/2$ on $B(\mathbb{Z}/2)$ and claimed $[\mathfrak{g}]$ was the generator of $H^2(\mathbb{Z}/2;\mathbb{Z}/2)$. This is wrong twice over. First, an abelian fibre carries $[\mathfrak{g}] = 0$ by Remark 11 ($\mathrm{Inn}K = 1$, ρ trivial, no lifting gerbe): the group $H^2(\mathbb{Z}/2;\mathbb{Z}/2) = \mathbb{Z}/2$ is nonzero, but the *inventory class* in it is zero. Second, the earlier degree-genuineness hypothesis "$N(\mathcal{C})$ homotopy 1-truncated" is itself refuted by this very space: $B(\mathbb{Z}/2) = K(\mathbb{Z}/2,1)$ *is* 1-truncated, yet $H^2(\mathbb{Z}/2;\mathbb{Z}/2) \neq 0$. The correct hypothesis in (i) is homotopy 1-*dimensionality*. The genuine minimal witness is the nonabelian D_4 of (ii).

Thus $[\mathfrak{g}]$ lives at cohomological degree 2 (nerve level N_3), vanishing on every 1-dimensional chain yet already nonzero on a single reversible loop carrying a nonabelian fibre. For the record, $H^2((\mathbb{Z}/2)^2;\mathbb{Z}/2) = \mathbb{F}_2^3$: $(\mathbb{Z}/2)^2$ hosts three independent $\mathbb{Z}/2$-gerbes — the D_4 and Q_8 classes and the $\mathbb{Z}/4 \times \mathbb{Z}/2$ class — of which the first were used in Remark 14. (As quadratic forms on $(\mathbb{Z}/2)^2$ there is a single Q_8 class x^2+xy+y^2 and three D_4 classes; a choice of three representatives spanning $\mathbb{F}_2^3$ is $x^2,\ x^2+xy,\ x^2+xy+y^2$ [6].)

7 Independence from the Zappa–Szép weld

Theorem 24 (Theorem C)**.** *Over the family of pairs (inventory datum, weld datum) on a fixed category $\mathcal{C}$, the gerbe class $[\mathfrak{g}] \in H^2(N(\mathcal{C});Z)$ and the Zappa–Szép weld $[\omega] \in H^2(\mathrm{Sk}_{\mathcal{C}};\mathcal{D})$ are logically independent: all four vanishing patterns occur, including $[\mathfrak{g}] = [\omega] \neq 0$. The mechanism is a difference of* input data*, not of coefficients: $[\mathfrak{g}]$ depends on the inventory fibre F, which is external to $\mathcal{C}$, while $[\omega]$ depends only on $\mathcal{C}$ and a candidate factorisation.*

The quantifier matters, and the earlier "different coefficient systems" explanation was false, so we state the mechanism carefully before proving it. On *some* categories the two classes live in *different* groups (different sites, different coefficients); on others — notably $\mathcal{C} = B(\mathbb{Z}/2)$ — they live in the *same* group $H^2(\mathbb{Z}/2;\mathbb{Z}/2) = \mathbb{Z}/2$ and can be equal. Equal coefficients therefore cannot be what makes them independent. What makes them independent is that they are functions of *different arguments*: fixing $\mathcal{C}$ and varying the inventory fibre moves $[\mathfrak{g}]$ while leaving $[\omega]$ fixed, and varying the factorisation datum moves $[\omega]$ while leaving $[\mathfrak{g}]$ fixed. We exhibit this on a single $\mathcal{C}$.

Proof. Work on $\mathcal{C} = B(\mathbb{Z}/2)$, where both classes can live in $H^2(\mathbb{Z}/2;\mathbb{Z}/2) = \mathbb{Z}/2$. The inventory datum is a fibre F; take $F \in \{D_4\text{-ring},\ \mathrm{pt}\}$, giving $[\mathfrak{g}] \neq 0$ (Theorem 22(ii)) resp. $[\mathfrak{g}] = 0$ ($\mathrm{Aut}F = 1$). The weld datum is a candidate Zappa–Szép factorisation of $\mathcal{C}$; take the re-entrant weld of [4], whose class $[\omega] = \varepsilon$ is the generator of $H^2(\mathbb{Z}/2;\mathbb{Z}/2)$, resp. the trivial factorisation $\mathcal{D} = 0$ with $[\omega] = 0$. These two binary choices are independent — one is a function of F, the other of the factorisation — so all four combinations are realised:

on $\mathcal{C} = B(\mathbb{Z}/2)$	weld = trivial ($[\omega] = 0$)	weld = re-entrant ($[\omega] \neq 0$)
fibre = pt ($[\mathfrak{g}] = 0$)	$(0,0)$	$(0, \neq 0)$
fibre = D_4-ring ($[\mathfrak{g}] \neq 0$)	$(\neq 0, 0)$	$(\neq 0, \neq 0),\ [\mathfrak{g}] = [\omega]$

In the bottom-right cell the two classes are *equal* (both the generator of the same group), yet they remain independent as functions: perturbing the fibre to pt kills $[\mathfrak{g}]$ without touching $[\omega]$, and vice versa. Hence neither vanishing implies the other. □

Remark 25 (Not a coefficient discriminator)**.** The earlier claim that "the coefficient system is the only discriminator" and that "there is no natural map comparing $H^2(N(\mathcal{C}); Z)$ with $H^2(\mathrm{Sk}_{\mathcal{C}}; \mathcal{D})$" is withdrawn: on $B(\mathbb{Z}/2)$ the two *are* the same group, with the same normalised cocycles, and the comparison map is the identity. The honest statement is the input-data one above. This is why, within the program, we describe $[\mathfrak{g}]$ as distinguished by its *input datum* (the inventory fibre) and not by a coefficient-system discriminator.

Remark 26 (Relation to tier one needs witnesses, not a degree count)**.** It is tempting to argue that the tier-one class $[\theta_R] \in H^1$ and the tier-two class $[\mathfrak{g}] \in H^2$ are "immediately independent because they differ in degree". That inference is invalid: classes of different degree can be functionally dependent (Bocksteins, cup squares, and d_3-type differentials all link classes across degrees). Independence of $[\theta_R]$ and $[\mathfrak{g}]$ must be established by witnesses, exactly as for Theorem 24. The full inventory obstruction is a two-tier filtration with associated graded $H^1(N(\mathcal{C}); R)$ (counts) and $H^2(N(\mathcal{C}); Z)$ (relabellings); whether the tiers split or form a Postnikov-type extension (a d_3 coupling $H^1 \to H^3$) is not determined here (Section 9).

8 When the gerbe is nonvacuous, and fungibility is doubly vacuous

A nonzero gerbe class is a genuine — and slightly surprising — constraint on *which* fibres carry a tier-two obstruction, and the band adds a parallel constraint. Together they make plain fungibility invisible to both.

Theorem 27 (Theorem D)**.** (i) $[\mathfrak{g}]$ *can be nonzero only if* $K = \mathrm{Aut}F$ *is* nonabelian with nontrivial centre*:* $\mathrm{Inn}K \neq 1$ *(so* ρ *can be nontrivial) and* $Z \neq 1$ *(so the coefficients are nonzero). The minimal such* K *are* D_4 *and* Q_8*.* (ii) *For* F *a bare* n*-element set of indistinguishable lots,* $\mathrm{Aut}F = S_n$*: fungible stock carries* neither *stratum for all* $n \geq 3$*,* $n \neq 6$*, since* S_n *is centreless (*$Z = 1 \Rightarrow [\mathfrak{g}] = 0$*) and complete (*$\mathrm{Out}S_n = 1 \Rightarrow \beta = 0$*). The low-order exceptions are disjoint:* $n = 6$ *is the lone band exception (*$\mathrm{Out}S_6 = \mathbb{Z}/2$*) and* $n = 2$ *the lone centre case (*$S_2 = \mathbb{Z}/2$*) — but* S_2 *is abelian, so even there* $[\mathfrak{g}] = 0$ *by (i).*

Proof. (i) If K is abelian then $\mathrm{Inn}K = 1$, ρ is trivial, and the gerbe of lifts is trivial, so $[\mathfrak{g}] = 0$ (Remark 11). If $Z = 1$ then $H^2(N(\mathcal{C}); Z) = 0$, so $[\mathfrak{g}] = 0$. Thus a nonzero class requires both $\mathrm{Inn}K \neq 1$ and $Z \neq 1$, i.e. K nonabelian with nontrivial centre; the smallest are D_4, Q_8 (order 8). (ii) S_n is complete for $n \neq 2, 6$ (centreless with all automorphisms inner, so Out $= 1$) and $\mathrm{Out}S_6 = \mathbb{Z}/2$ (Rotman [14], Thms. 7.5–7.7); $Z(S_n) = 1$ for $n \geq 3$, $= \mathbb{Z}/2$ for $n = 2$, $= 1$ for $n \leq 1$. Feed these into (i) and Theorem 17. For $n = 2$, $S_2 = \mathbb{Z}/2$ is abelian, so although the *group* $H^2(\mathbb{Z}/2; \mathbb{Z}/2)$ is the home of a nonzero class, the *inventory* class is 0 by (i). The exceptional values $n \in \{2, 6\}$ are distinct and each triggers at most one stratum's coefficient group, as tabulated (`scratch/band_boundary_verify.py`, `out_s6.py`). □

8.1 The sharp criterion for the n-lot ring

Theorem 27(i) is a *necessary* condition only, and it must not be read as a biconditional. An earlier registry entry and cover email did read it as one — "$[\mathfrak{g}]$ is essentially nonzero $\iff$ K is nonabelian with nontrivial centre" — and that biconditional is **false**. We retract it and record the sharp replacement, which for the natural inventory fibre ("n fungible lots arranged in a ring") is a clean arithmetic line.

Call the inventory gerbe *essentially nonzero* if $[\mathfrak{g}] \neq 0$ for *some* base $\mathcal{C}$ and inner transport $\rho : \mathcal{C} \to B(\mathrm{Inn}K)$ — equivalently, if strict global inventory with inner comparison is obstructed for some supply chain.

Theorem 28 (Sharp criterion for the n-lot ring)**.** *Let the fibre be n lots in a ring, so $K = \mathrm{Aut}F = D_n$ is dihedral of order $2n$ ($n \geq 3$). Assume trivial comparison data, $k_{g,f} \equiv e$ (§2.3). Then, over inner transports (the trivial-band stratum $\beta = 0$),*

$$[\mathfrak{g}] \text{ essentially nonzero} \iff (\star) \text{ non-split} \iff [\star] \neq 0 \in H^2(\mathrm{Inn}K; Z) \iff 4 \mid n.$$

Equivalently, over the trivial-band stratum with $k \equiv e$, strict global inventory on the n-lot ring is always *rectifiable unless $4 \mid n$. (A nontrivial band can obstruct inventory even when $4 \nmid n$, and a nontrivial central k moves the line to $2 \mid n$: Remark 30.)*

Proof. The first equivalence is the universal-base argument: ($\Leftarrow$) take $\mathcal{C} = B(\mathrm{Inn}K)$, $\rho = \mathrm{id}$, whence $[\mathfrak{g}] = [\star]$; ($\Rightarrow$) $[\mathfrak{g}] = \rho^*[\star]$ vanishes for every $(\mathcal{C}, \rho)$ once $[\star] = 0$, since pullback is a homomorphism. The second equivalence is the classical correspondence between central extensions and H^2 already used in Lemma 9 and Theorem 12: $[\star] = 0$ iff $(\star)$ splits (the coefficients are untwisted, as $\mathrm{Inn}K$ acts trivially on Z). For the arithmetic, $Z(D_n) = 1$ for n odd and $\{1, r^{n/2}\} \cong \mathbb{Z}/2$ for n even, while $\mathrm{Inn}D_n \cong D_n$ (n odd) resp. $D_{n/2}$ (n even). Writing $n = 2m$ in the even case, a complement to the central $\mathbb{Z}/2$ in D_{2m} would exhibit $D_{2m} \cong \mathbb{Z}/2 \times D_m$; a centre count settles this, since $|Z(D_{2m})| = 2$ whereas $|Z(\mathbb{Z}/2 \times D_m)| = 2\,|Z(D_m)|$ equals 2 iff m is odd. Hence $(\star)$ splits iff $m = n/2$ is odd, i.e. iff $4 \nmid n$; for n odd, $Z = 1$ and $(\star)$ splits vacuously. The full argument (and an explicit D_m complement when $4 \nmid n$) is in the companion note [7] (own-verification; the split test is computed for $n = 3..16$, with non-split occurring exactly at $n \in \{4, 8, 12, 16\}$). □

Remark 29 (Why "nonabelian with centre" is not sufficient: the six-lot ring)**.** The six-lot ring has $K = D_6 \cong \mathbb{Z}/2 \times S_3$: nonabelian, with centre $\mathbb{Z}/2 \neq 1$, so it *satisfies* Theorem 27(i). Yet $6 = 2 \cdot 3$ with $n/2 = 3$ odd, so $(\star)$ splits (explicit complement $\langle r^2, s\rangle \cong S_3$), $[\star] = 0$, and $[\mathfrak{g}] = 0$ for *every* base $\mathcal{C}$ and *inner* transport ρ: the six-lot inventory is always rectifiable over the trivial band (the banded case is a genuine exception, Remark 30). Thus D_6 is the honest witness that the *necessary* condition of Theorem 27(i) is not *sufficient*; non-splitting of $(\star)$ — the condition $4 \mid n$ — is the sharp line. The four-lot witness $K = D_4$ of Theorem 22(ii) is the minimal case $4 \mid n$.

Remark 30 (The sharp line is trivial-band; a nontrivial band obstructs independently)**.** Theorem 28 classifies the *trivial-band* stratum ($\beta = 0$, inner transports): "always rectifiable unless $4 \mid n$" is a statement about that stratum, not about every supply chain. A nontrivial band carries its own obstruction, on top of the central one, and can block inventory even when $4 \nmid n$. The six-lot ring is again the honest witness. Although $4 \nmid 6$ makes $(\star)$ split, so that the *central* obstruction vanishes, the base $\mathcal{C} = B(\mathbb{Z}/2)$ with the *outer* involution of D_6 given by $r \mapsto r^5$, $s \mapsto sr$ (a nontrivial element of $\mathrm{Out}D_6 \cong \mathbb{Z}/2$) admits *no* global inventory: the band does not even vanish, so the degree-two coefficients never untwist. Thus "always rectifiable unless $4 \mid n$" holds only for the trivial band; the banded strata are governed by the forced-correlation analysis of the companion note [8], where the two ends of a T^2 base are shown to couple. The applications headline below is accordingly stated for the trivial-band, $k \equiv e$ regime.

n	$K = D_n$	Z	$\mathrm{Inn}K$	$(\star)$	$[\mathfrak{g}]$ ess. nonzero?
3	S_3	1	S_3	split (vacuous)	no
4	D_4	$\mathbb{Z}/2$	$(\mathbb{Z}/2)^2$	**non-split**	**yes**
5	D_5	1	D_5	split (vacuous)	no
6	$D_6 = \mathbb{Z}/2 \times S_3$	$\mathbb{Z}/2$	S_3	split	no
8	D_8	$\mathbb{Z}/2$	D_4	**non-split**	**yes**
10	D_{10}	$\mathbb{Z}/2$	D_5	split	no
12	D_{12}	$\mathbb{Z}/2$	D_6	**non-split**	**yes**

The "yes" column is exactly $\{n : 4 \mid n\}$. The criterion sharpens the applications story: among gauge-structured stock, whether a fibre carries a tier-two obstruction is decided not by the coarse "nonabelian with centre" test but by a non-splitting (equivalently, a divisibility) condition on its relabelling group.

The applications boundary is sharp. The nonabelian inventory phenomenon is *invisible to fungibility*: ≥ 3 interchangeable units create no degree-two relabelling obstruction (S_n centreless) and no band obstruction (S_n complete), except the isolated S_6 curiosity. Both nontrivial strata require *gauge/torsor-structured* stock — a fibre whose automorphism group is nonabelian with nontrivial centre (tier-2) or nontrivial outer group (band). This is the economically meaningful regime: allocation defined up to a gauge group K (so $\mathrm{Aut}F = K$ acts by translation, $Z = Z(K)$, Out measuring how the process twists the gauge), not plain interchangeability. The grant headline is exactly this: *fungible inventory carries no nonabelian obstruction; the phenomenon needs structured assets, not commodities.*

9 Scope, open problem, and the two-tier bottom line

Refining inventory comparison from *equality of counts* to *isomorphism of stock-states* raises the consistency obstruction by exactly one categorical level, through a staged obstruction. A global inventory exists iff the band $\beta \in H^1(N(\mathcal{C}); \mathrm{Out}(\mathrm{Aut}F))$ vanishes *and* then the gerbe class $[\mathfrak{g}] \in H^2(N(\mathcal{C}); Z(\mathrm{Aut}F))$ vanishes (Corollary 18), the latter being the classical Schreier / Eilenberg–MacLane central-extension lifting criterion read as inventory rectification (Theorem 12, Remark 13). The gerbe class is genuinely degree-two, vanishing on every 1-dimensional chain but nonzero already on a single reversible loop $B(\mathbb{Z}/2)$ carrying a nonabelian four-lot ring D_4 (Theorem 22); it is independent of the weld $[\omega]$ by a difference of *input data*, not coefficients — the two can even coincide on $B(\mathbb{Z}/2)$ (Theorem 24); and both strata are invisible to plain fungibility, requiring gauge/torsor-structured stock (Theorem 27). Within the container program this is the fourth distinct H^2 home on DCont $\simeq$ Cat, after Zappa–Szép holonomy, Ferri prunability, and the Baues–Wirsching class — distinguished by its *input datum*.

The honest open problem. Proposition 21 witnesses a nonvacuous band but does *not* classify nontrivial-band gerbes; that is Giraud's full nonabelian machinery (the torsor over the twisted $H^2(N(\mathcal{C}); Z)_\beta$), out of scope here. The natural next target is the *coupled* stratum, where band and centre interact: a base with $H^2 \neq 0$ and a nontrivial outer transport, e.g. $\mathcal{C} = B(\mathbb{Z}^2)$ ($N \simeq T^2$) with $K = (\mathbb{Z}/p)^2$ and a commuting outer pair (A, B), computing the twisted differential d_2 coupling β to the centre class. We do not have this; it is recorded as the PROVE/LEAN follow-up (a bar-complex oracle on T^2). Relatedly (Remark 26), the coupling between tiers one and two — split or Postnikov extension — is stated as a filtration but not computed.

For the grant. The worked economic example to develop is the torsor-fibre case $\mathrm{Aut}F = K$ acting on itself, with $Z = Z(K)$ — gauge-valued allocation, where stock is defined only up to a group action and the tier-two obstruction has real content. The applications payoff of Theorem 27 is that the economically relevant non-rectifiable inventory is not the many-interchangeable-units case but the gauge-structured one; and the band supplies a strictly prior, genuinely nonabelian invariant of the same situation.

References

[1] D. Ahman, T. Uustalu, *Directed containers as categories*, Proc. MSFP 2016, EPTCS 207, 2016, 89–98; arXiv:1604.01187.

[2] MacBeth, *Supply-chain inventory consistency is a degree-one class, and is independent of the Zappa–Szép weld obstruction* (Kodamai internal note, 6 October 2026; proved).

[3] MacBeth, *The global-closure obstruction* (G) *is a cohomology class* (Kodamai internal note, 11 June 2026; proved, Lean-adjacent).

[4] MacBeth, *The re-entrancy obstruction is the token-mutation bit: an analytic proof for orchestration Zappa–Szép products* (Kodamai internal note, 20 July 2026; proved, $[\omega] = \varepsilon$ core Lean-verified).

[5] MacBeth, *Two* $[\omega]$*-sites are irreducibly distinct* (Kodamai internal note, 14 August 2026; proved).

[6] MacBeth, *Reduced-nerve* H^2 *witness engine* (Lean 4; one engine, three machine-checked witnesses: $H^2(B(\mathbb{Z}/2)) = \mathbb{Z}/2$, the $(\mathbb{Z}/2)^2$ diamond basis, the D_4 non-splitting), Kodamai internal Lean note, 9 October 2026.

[7] MacBeth, *The sharp criterion for strict global inventory on the n-lot ring: the gerbe fails exactly when* $4 \mid n$ (Kodamai internal note, 11 October 2026; proved, own-verification, computation-corroborated).

[8] MacBeth, *The inventory gerbe couples on the torus: forced correlation of the two* T^2 *ends* (Kodamai work-in-progress note, 2026).

[9] J. Giraud, *Cohomologie non abélienne*, Grundlehren der mathematischen Wissenschaften **179**, Springer, 1971.

[10] L. Breen, *Bitorseurs et cohomologie non abélienne*, in *The Grothendieck Festschrift, Vol. I*, Progr. Math. **86**, Birkhäuser, 1990, 401–476.

[11] G. Hoff, *Cohomologies et extensions de catégories*, Math. Scand. **74** (1994), 191–207, doi:10.7146/math.scand.a-12489; see also *On the cohomology of categories*, Comm. Fac. Sci. Univ. Ankara Sér. A1 **22** (1973), 175–197.

[12] C. Wells, *Extension theories for categories* (unpublished preprint, Case Western Reserve University, 1979); see also D. Bourn, *The shift functor and the comprehensive factorization for internal groupoids*, Cahiers Topologie Géom. Différ. Catég. **28** (1987), 197–226.

[13] H.-J. Baues, G. Wirsching, *Cohomology of small categories*, J. Pure Appl. Algebra **38** (1985), 187–211.

[14] J. J. Rotman, *An Introduction to the Theory of Groups*, 4th ed., Graduate Texts in Mathematics **148**, Springer, 1995 (Thms. 7.5–7.7: S_n complete for $n \neq 2, 6$; $\mathrm{Out}S_6 = \mathbb{Z}/2$).

[15] J. C. Baez, A. D. Lauda, *Higher-dimensional algebra V: 2-groups*, Theory Appl. Categ. **12** (2004), 423–491.

[16] N. Johnson, D. Yau, *2-Dimensional Categories*, Oxford University Press, 2021.

[17] C. A. Weibel, *An Introduction to Homological Algebra*, Cambridge Studies in Advanced Mathematics **38**, Cambridge Univ. Press, 1994.